

Niche breadth as a driver of SDM performance: a comparison of MaxEnt, GLM and BRT using virtual species

Do MaxEnt, GLM and BRT differ systematically in their predictive performance when modelling virtual species with narrow, intermediate and broad ecological niches, and which algorithm performs best for which niche breadth?

Introduction

The ecological niche concept is central to the understanding of species distributions, and species distribution models (SDMs) have become the primary tool for spatial predictions of species occurrence (Elith & Leathwick 2009). Species can be classified as habitat generalists or specialists depending on their ability to adapt to environmental variables, thereby influencing the modelling output (Ainsworth & Drake 2020, p. 2f.; Connor et al. 2018, p. 1271).

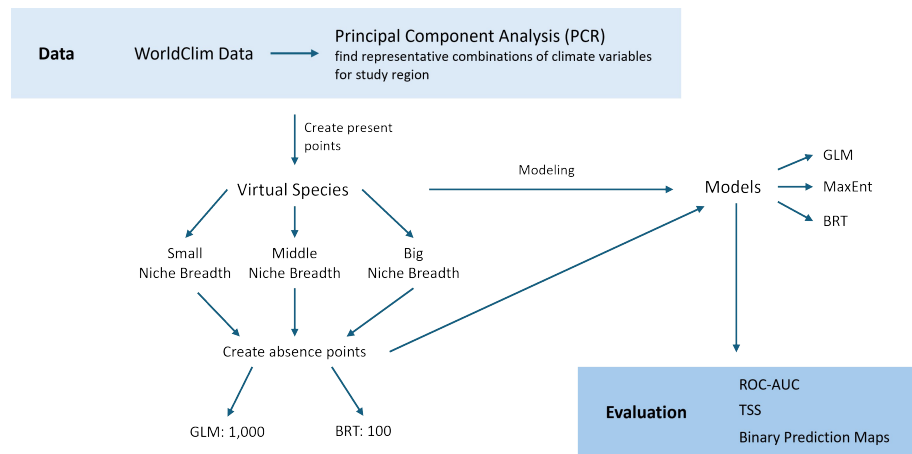
Since existing studies report contradictory findings on the relationship between niche breadth and SDM performance (Connor et al. 2018; Tassarolo et al. 2021, p. 2), a controlled comparison using virtual species is needed to disentangle the effect of niche breadth from confounding factors inherent to real occurrence data. In order to investigate the influence of niche width as a determinant of model performance, we will use virtual species in this study, as these are species whose entire niche and range are known (Saupe et al. 2012, p. 12). We will use three different algorithms (GLM, MaxEnt, BRT) to model the distribution of these species in Germany.

Data and Methods

Three virtual species with contrasting niche breadths will be generated based on bioclimatic variables from the WorldClim 2.1 dataset (Fick & Hijmans 2017), available at a spatial resolution of 30 arc-seconds (~1 km) for Germany. Annual mean temperature, altitude, seasonal precipitation and annual mean evapotranspiration are the environmental variables who have a direct influence on the ecology niche from species (Soultan & Safi 2017, p. 3). A Principal Component Analysis (PCA) (Leroy et al., p. 600) was conducted on four bioclimatic variables (BIO1: Annual Mean Temperature; BIO4: Temperature Seasonality; BIO12: Annual Precipitation; BIO15: Precipitation Seasonality). The first two principal components (PC1: 41.8 %, PC2: 40.2 %) together explained 82 % of the total environmental variance across Germany and will be used to define the ecological niche of each virtual species. With this variables the virtual species will be generated using the R package *virtualspecies* (Leroy et al. 2016). The Niche breadth is controlled by the mean standard deviation (σ) of a Gaussian response function along both PC axes, representing the proportion of the total environmental range covered by each species: broad-niche species (generalist) $\sigma = 0.8$ (80 %), intermediate-niche species $\sigma = 0.5$ (50 %), and narrow-niche species (specialist) $\sigma = 0.2$ (20 %) (cf. Soultan & Safi 2017, p. 3). To ensure comparability of model performance across niche breadths, an equal number of presence points ($n = 100$) will be sampled for each virtual species, as sample size is known to strongly influence SDM performance independently of niche characteristics (Hernandez et al. 2006). Pseudo-absence selection was tailored to each algorithm: GLM used 1,000 randomly selected pseudo-absences with equal weighting, while BRT used pseudo-absences equal in number to the available presences ($n = 100$), averaged across 10 runs (Barbet-Massin et al. 2012). MaxEnt does not require pseudo-absences as it is a presence-only method (Phillips et al. 2006).

The models will be evaluated using threshold-dependent and threshold-independent metrics, to offer a range of evaluations. TSS is threshold-dependent and measures the model performance. It is based on the sensitivity and specificity, with a great advantage of it not being dependent of prevalence (Allouche et al. 2006, 1224f.;

Connor et al. 2018, 1271). The area under the ROC curve (AUC) also measures the performance, but different to TSS is threshold-independent (Allouche et al. 2006, 1224). This metric assesses whether presence locations receive higher predicted probability values than absence locations (McPherson & Jetz 2007, 142).



Discussion

The niche breadth is expected to have a strong influence on the performance of the different model. Other studies have discovered a correlation between specialized species and a higher accuracy, with descending performances the larger the niche (Connor et al., 2018, p. 1275; Hernandez et al., 2006, p. 774), that we also expect in our analysis. Additionally, we put forward the following hypotheses:

H1: Specialist species with narrow niches are modelled more accurately than generalists across all three algorithms (Connor et al. 2018; Hernandez et al., 2006; Tessarolo et al. 2021).

H2: MaxEnt and BRT outperform GLM across all niche breadths, but may differ from each other in their sensitivity to niche breadth (Elith et al. 2008; Valavi et al. 2022).

One limiting factor is the small number of species and algorithms as well as the few evaluation metrics used, leading to restricted results with limited informative value. Another factor that could be implemented in future research is the number of sample sizes of the different species. As Hernandez et al. (2006) have discussed, this also has an important influence on the performance of the models and should therefore be considered. Furthermore, the virtual species were simulated based on a static climatic time stamp, including seasonality variables (BIO4, BIO15) that describe spatial climatic patterns rather than temporal dynamics in this case. Because of that, the transferability of results to real species responding to actual seasonal variation may be limited.

Timetable

17.06.2026	Project outline
30.06.2026	Write code
12.07.2026	Data analysis + Discussion
31.07.2026	Final writing —> finished report
20.08.2026	Buffer time

References

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